

A Catalan Classification of Extremal Even-Intersecting Lattice Paths

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Abstract

Shankar proved that a family of North-East lattice paths from $(0,0)$ to (n,n) in which every two distinct paths have an even number of common edges has size at most 2^n . He constructed C_n extremal families from the noncrossing perfect matchings of $[2n]$ and conjectured that these are all the extremal families. This note proves the conjecture. The main step is an equality classification for the quadratic lemma underlying Shankar's upper bound: its equality cases are indexed by Dyck prefixes. Two auxiliary rigidity statements, one for these Dyck-prefix partitions and one for the parity kernel on the Boolean cube, then determine the extremal relation completely.

Keywords. lattice paths, extremal set theory, Eventown, Catalan numbers, noncrossing matchings.

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1 Introduction

A North-East lattice path from $(0,0)$ to (n,n) is a word in $\{E, N\}^{2n}$ with n copies of each letter. Two such paths are *even-intersecting* if the number of common lattice edges is even. Shankar [1] proved that every pairwise even-intersecting family has size at most 2^n . For a noncrossing perfect matching M of $[2n]$, he also considered

$$\mathcal{F}_M = \{w \in \{E, N\}^{2n} : \{w_i, w_j\} = \{E, N\} \text{ for every } \{i, j\} \in M\}. \quad (1)$$

Each $\mathcal{F}_M$ has size 2^n and is pairwise even-intersecting [1, Section 4]. Since the noncrossing perfect matchings of $[2n]$ are counted by the Catalan number C_n (see, for example, [2]), this gives at least C_n extremal families. Shankar conjectured that there are no others [1, Conjecture 4.1] and verified the conjecture for $n \leq 5$ [1, Section 4].

The purpose of this note is to prove the conjecture in its stronger structural form.

Theorem 1.1. *Let $n \geq 1$, and let $\mathcal{F}$ be a family of lattice paths from $(0,0)$ to (n,n) such that every two distinct members have an even number of common edges. If $|\mathcal{F}| = 2^n$, then there is a unique noncrossing perfect matching M of $[2n]$ for which*

$$\mathcal{F} = \mathcal{F}_M.$$

Consequently there are exactly C_n extremal families.

The proof follows the equality structure in Shankar's argument rather than introducing a different upper bound. It has three ingredients. First, equality in the quadratic lemma from [1] is classified by Dyck prefixes. Second, two such equality partitions cannot be cross-compatible unless their Dyck prefixes have the same matched edges. Third, a weight-preserving automorphism of the Boolean cube that preserves the path-intersection parity kernel is trivial. The essential point is to retain the full equality information throughout the induction.

2 The parity kernel and Dyck-prefix partitions

Identify an m -step path with a word $x = x_1 \cdots x_m \in \{0, 1\}^m$, where $1 = N$ and $0 = E$. Write $\text{wt}(x) = \sum_i x_i$. Following [1], put

$$B_m(x, y) = |E(x) \cap E(y)| \pmod{2} \in \mathbb{F}_2.$$

Equivalently,

$$B_m(x, y) = \sum_{i=1}^m \mathbf{1}_{\{x_i=y_i\}} \mathbf{1}_{\{\sum_{j<i} x_j = \sum_{j<i} y_j\}} \quad \text{in } \mathbb{F}_2. \quad (2)$$

In particular $B_m(x, x) = m \pmod{2}$.

The last coordinate gives the basic recurrence. If $p, q \in \{0, 1\}^{m-1}$ and $\alpha, \beta \in \{0, 1\}$, then

$$B_m(p\alpha, q\beta) = B_{m-1}(p, q) + \mathbf{1}_{\{\alpha=\beta\}} \mathbf{1}_{\{\text{wt}(p)=\text{wt}(q)\}}. \quad (3)$$

2.1 Dyck-prefix blocks

A *Dyck prefix* of length m is a word $P = P_1 \cdots P_m$ in the alphabet $\{U, D\}$ whose height never becomes negative. Match every D to the most recent unmatched U . Let $\mathcal{E}(P)$ be the resulting noncrossing partial matching on $[m]$, and let

$$u_1 < \cdots < u_r$$

be the unmatched positions. If $k = |\mathcal{E}(P)|$, then $m = 2k + r$.

For $\varepsilon = (\varepsilon_1, \dots, \varepsilon_r) \in \{0, 1\}^r$ define

$$\mathcal{A}_\varepsilon(P) = \{x \in \{0, 1\}^m : x_a \neq x_b \text{ for every } \{a, b\} \in \mathcal{E}(P), \quad x_{u_i} = \varepsilon_i \ (1 \leq i \leq r)\}. \quad (4)$$

Thus every block has size 2^k and lies in the weight level $k + \text{wt}(\varepsilon)$. Put

$$\Omega(P) = \bigcup_{\varepsilon \in \{0, 1\}^r} \mathcal{A}_\varepsilon(P).$$

The next cancellation fact is the reason these blocks appear naturally.

Lemma 2.1 (cancellation). *Let P be a Dyck prefix of length m with r unmatched positions. Suppose $x, y \in \Omega(P)$, and delete from both words every coordinate belonging to an edge of $\mathcal{E}(P)$. If the resulting words are $\bar{x}, \bar{y} \in \{0, 1\}^r$, then*

$$B_m(x, y) = B_r(\bar{x}, \bar{y}). \quad (5)$$

In particular, if $x \in \mathcal{A}_\varepsilon(P)$ and $y \in \mathcal{A}_\eta(P)$, then

$$B_m(x, y) = B_r(\varepsilon, \eta). \quad (6)$$

Proof. No unmatched position of a Dyck prefix can lie strictly inside a matched edge: otherwise it would still be on top of the stack when that edge closed. Hence every nonempty matching $\mathcal{E}(P)$ contains an innermost edge $\{i, i+1\}$. Both x and y use one 0 and one 1 at these two positions. If the orientations agree, the two positions contribute either two common edges or none; if the orientations disagree, they contribute none. Hence their contribution is even. After the two positions are deleted, both paths have been translated by the same vector $(1, 1)$ from that point onward, so all later common-edge incidences are unchanged. Deleting innermost edges successively proves (5). $\square$

It follows at once that the blocks (4) satisfy the three compatibility conditions that will occur in Section 3: within a block the value of B_m is $r \equiv m \pmod{2}$, while on a product of two blocks it is constant.

We shall also need the following two-step test.

Lemma 2.2 (square test). *Let $s \geq 1$ and $y = y'\beta\gamma \in \{0,1\}^{s+1}$ with $y' \in \{0,1\}^{s-1}$. The following are equivalent:*

- (i) $B_{s+1}(z10, y) = B_{s+1}(z01, y)$ for every $z \in \{0,1\}^{s-1}$;
- (ii) $\beta \neq \gamma$.

Proof. Let $h = \text{wt}(z)$ and $q = \text{wt}(y')$. Applying (3) twice gives

$$\begin{aligned} B_{s+1}(z10, y'\beta\gamma) + B_{s+1}(z01, y'\beta\gamma) \\ = \mathbf{1}_{\{h=q\}} + \mathbf{1}_{\{\gamma=0\}}\mathbf{1}_{\{h=q+\beta-1\}} + \mathbf{1}_{\{\gamma=1\}}\mathbf{1}_{\{h=q+\beta\}}. \end{aligned}$$

If $\beta \neq \gamma$, the right-hand side is identically zero. If $\beta = \gamma = 0$, take $h = q$; if $\beta = \gamma = 1$, again take $h = q$. Since $0 \leq q \leq s-1$, such a word z exists, and the displayed value is then 1. $\square$

2.2 Cross-rigidity

For Dyck prefixes P, Q of the same length, write $P \preccurlyeq Q$ if, for every block A of the partition $\{\mathcal{A}_\varepsilon(P)\}$, every $x, x' \in A$, and every $y \in \Omega(Q)$,

$$B_m(x, y) = B_m(x', y). \quad (7)$$

Thus changing the orientations of the matched edges of P is invisible to every word satisfying the matching constraints of Q .

Lemma 2.3. *For Dyck prefixes P, Q of length m ,*

$$P \preccurlyeq Q \implies \mathcal{E}(P) \subseteq \mathcal{E}(Q).$$

Proof. Proceed by induction on m . The case $m = 0$ is immediate. Assume $m \geq 1$, and remove the last symbol from P and Q , obtaining Dyck prefixes P^- and Q^- . Every word in $\Omega(Q^-)$ can be extended to a word in $\Omega(Q)$: if the last symbol of Q is U , choose either last bit; if it is D , the last bit is forced to be opposite to the bit at its matched opener. Likewise, any two words in the same block for P^- can be extended with a common last bit to two words in one block for P ; when the last symbol of P is D , the bit at its opener is fixed inside a block of P^- . Since all words in one block have the same weight, the correction term in (3) is the same for the two extensions. Hence

$$P^- \preccurlyeq Q^-.$$

By induction, $\mathcal{E}(P^-) \subseteq \mathcal{E}(Q^-)$.

If the last symbol of P is U , this already gives $\mathcal{E}(P) \subseteq \mathcal{E}(Q)$. Suppose instead that it is D , and let e be its matched edge. Delete the endpoints of all edges in $\mathcal{E}(P^-)$. By induction these edges also belong to $\mathcal{E}(Q^-)$. Lemma 2.1, applied successively to the same matched pairs on both sides of (7), shows that every defining B -equality is unchanged; hence $\preccurlyeq$ descends to the reduced prefixes. Removing the endpoints of a matched U – D pair preserves the Dyck-prefix property. On the P -side this is equivalently the cancellation associated with P^-U , whose matching is exactly $\mathcal{E}(P^-)$. The reduced prefix coming from P has the form

$$U^s D$$

for some $s \geq 1$: all surviving vertices before the final D are unmatched, and the last of them is its opener. Let Q' be the reduced prefix coming from Q .

A block of $U^s D$ is $\{z10, z01\}$ for some $z \in \{0, 1\}^{s-1}$. Thus, for every $y \in \Omega(Q')$,

$$B_{s+1}(z10, y) = B_{s+1}(z01, y) \quad \text{for all } z \in \{0, 1\}^{s-1}.$$

Lemma 2.2 gives $y_s \neq y_{s+1}$ for every $y \in \Omega(Q')$. In a Dyck-prefix matching, two positions have opposite bits in every word of $\Omega(Q')$ if and only if they are matched to one another: if they lie on different matching edges, or if one of them is unmatched, the relevant orientations and unmatched bits can be chosen independently to make the two bits equal. Therefore $\{s, s+1\} \in \mathcal{E}(Q')$. Lifting this edge back gives $e \in \mathcal{E}(Q)$. $\square$

Proposition 2.4 (cross-rigidity). *Let P, Q be Dyck prefixes of length m . Suppose that B_m is constant on the product of every block of the P -partition with every block of the Q -partition. Then $P = Q$.*

Proof. The hypothesis gives $P \preceq Q$. Since B_m is symmetric, it also gives $Q \preceq P$. Lemma 2.3 yields $\mathcal{E}(P) = \mathcal{E}(Q)$. A Dyck prefix is determined by its matching edges: the right endpoints are the D -positions, and every other position is a U -position. Hence $P = Q$. $\square$

3 Equality in the quadratic lemma

Shankar's proof of the even-intersection bound rests on the following inequality [1, Lemma 2.3]. Empty members do not affect the inequality and may be discarded. The equality classification is stated simultaneously.

Theorem 3.1 (quadratic equality classification). *Let $A_1, \dots, A_t$ be nonempty pairwise disjoint subsets of $\{0, 1\}^m$ such that*

- (i) *each A_i is contained in one Hamming-weight level;*
- (ii) *$B_m(x, y) = m \pmod{2}$ for all $x, y \in A_i$;*
- (iii) *for $i \neq j$, the function B_m is constant on $A_i \times A_j$.*

Then

$$\sum_{i=1}^t |A_i|^2 \leq 2^m. \tag{8}$$

Equality holds if and only if there is a Dyck prefix P of length m such that, up to reordering,

$$\{A_1, \dots, A_t\} = \{\mathcal{A}_\varepsilon(P) : \varepsilon \in \{0, 1\}^r\},$$

where r is the number of unmatched positions of P . The prefix P is unique.

The equality analysis reduces to the following local Boolean-cube classification. The same lemma will later control automorphisms.

For $x, y \in \{0, 1\}^d$, set

$$\Delta_{x,y}(z) = B_d(x, z) + B_d(y, z) \in \mathbb{F}_2.$$

Let $\mathcal{L}_j^{(d)} = \{z \in \{0, 1\}^d : \text{wt}(z) = j\}$, with the convention that this set is empty if $j \notin \{0, \dots, d\}$.

Lemma 3.2 (localized pairs). *Let $u, v \in \{0, 1\}^d$ satisfy $\text{wt}(u) = k + 1$ and $\text{wt}(v) = k$. Assume*

$$\Delta_{u,v}(z) = 0 \quad \text{whenever } \text{wt}(z) \notin \{k, k + 1\}. \quad (9)$$

Then either

$$u = t1, \quad v = t0 \quad (10)$$

for some $t \in \{0, 1\}^{d-1}$, or $d = 2k + 1 \geq 3$ and

$$u = 1(10)^k, \quad v = 0(01)^k. \quad (11)$$

For the exceptional pair,

$$\Delta_{u,v}(z) = 1 \iff \text{wt}(z) \in \{k, k + 1\}. \quad (12)$$

Proof. Write $u = p\alpha$, $v = q\beta$, and $z\gamma$ for the test word. From (3),

$$\Delta_{p\alpha, q\beta}(z\gamma) = \Delta_{p,q}(z) + \mathbf{1}_{\{\alpha=\gamma\}} \mathbf{1}_{\{\text{wt}(p)=\text{wt}(z)\}} + \mathbf{1}_{\{\beta=\gamma\}} \mathbf{1}_{\{\text{wt}(q)=\text{wt}(z)\}}. \quad (13)$$

The proof is a finite-state induction; the details below account for the exceptional alternating state.

First consider the following auxiliary states.

- $S_d(k)$: $\text{wt}(x) = \text{wt}(y) = k$ and $\text{supp } \Delta_{x,y} \subseteq \mathcal{L}_k^{(d)}$;
- $E_d(k)$: $\text{wt}(x) = \text{wt}(y) = k$ and $\Delta_{x,y} = \mathbf{1}_{\mathcal{L}_k^{(d)}}$;
- $L_d^-(k)$: $\text{wt}(x) = k + 1$, $\text{wt}(y) = k$, and $\text{supp } \Delta_{x,y} \subseteq \mathcal{L}_k^{(d)}$;
- $L_d^+(k)$: the same weight condition, with support contained in $\mathcal{L}_{k+1}^{(d)}$.

The case $d = 0$ is immediate. A simultaneous induction on d , using (13), gives

$$S_d(k) \implies x = y, \quad E_d(k), L_d^-(k), L_d^+(k) \text{ are impossible.} \quad (14)$$

The transition rules are as follows. A terminal pattern whose implied prefix weight lies outside $\{0, \dots, d - 1\}$ is infeasible; whenever a pattern is feasible, each weight level used in the argument below is nonempty. In the state $E_d(k)$, equal terminal bits are impossible. For terminal pattern 00, take a prefix word z of weight k : the two choices $z0, z1$ have target values 1, 0, whereas the left side of (13) is independent of γ . For terminal pattern 11, the same contradiction is obtained from a prefix word of weight $k - 1$. Unequal terminal bits force $\Delta_{p,q} \equiv 0$ for a pair of adjacent weights, contradicting the induction hypothesis for L^-/L^+ . For $L_d^-(k)$, the terminal patterns 00 and 01 give an immediate contradiction by evaluating (13) on a word of weight $k + 1$; the pattern 10 reduces to $E_{d-1}(k)$, while 11 reduces to $L_{d-1}^+(k - 1)$. If $\bar{x}$ denotes the bitwise complement of x , then $B_d(\bar{x}, \bar{z}) = B_d(x, z)$. Hence simultaneous complementation of the two words and the test word, followed by swapping the first two words, sends an L^+ state to an L^- state. Finally, in $S_d(k)$, equal terminal bits force $\Delta_{p,q} \equiv 0$, so induction gives $p = q$; unequal terminal bits again force an adjacent-weight pair with zero profile, contradicting L^-/L^+ .

It remains to track two alternating states. Let

$$\begin{aligned} M_d(k) : & \quad \text{wt}(x) = k + 1, \text{ wt}(y) = k - 1, \quad \text{supp } \Delta_{x,y} \subseteq \mathcal{L}_k^{(d)}, \\ T_d(k) : & \quad \text{wt}(x) = k + 1, \text{ wt}(y) = k, \quad \Delta_{x,y} = \mathbf{1}_{\mathcal{L}_k^{(d)} \cup \mathcal{L}_{k+1}^{(d)}}. \end{aligned}$$

The remaining two states have the following transitions, again by (13). In an $M_d(k)$ state, the patterns 00 and 01 are contradicted by testing a word of weight $k+1$, while 11 is contradicted by testing a word of weight $k-2$ (the pattern is feasible only when that layer is nonempty). The only surviving pattern is 10, and the prefix then has value 1 on exactly the two layers $k-1, k$; hence it is a $T_{d-1}(k-1)$ pair. In a $T_d(k)$ state with $d > 1$, the patterns 00 and 11 are contradicted by testing a prefix word of weight k . For the pattern 10, the two last-bit choices give the same value. If $k > 0$, a prefix word of weight $k-1$ gives target values 0, 1; if $k = 0$, a prefix word of weight 1 gives target values 1, 0 (and this level is nonempty because $d > 1$). Thus only 01 survives, and (13) says that its prefix has support exactly on layer k , so it is an $M_{d-1}(k)$ pair. Since $T_1(0)$ consists only of $(1, 0)$, induction gives

$$M_{2k}(k) : (x, y) = (11(01)^{k-1}, 00(10)^{k-1}) \quad (k \geq 1), \quad (15)$$

$$T_{2k+1}(k) : (x, y) = (1(10)^k, 0(01)^k). \quad (16)$$

No M or T state occurs in the opposite parity.

Now apply (13) to a pair satisfying (9). According to the terminal pattern (α, β) , its prefix lies in the following state:

(α, β)	prefix state	consequence
00	L^-	impossible
11	L^+	impossible
10	S	$p = q$
01	M	the alternating pair (11).

The 10 case is exactly (10). The 01 case, together with (15), yields (11); its profile is (12) by (16). When $d = 1$, this alternating pair is the canonical pair itself, so the exceptional alternative is new only for $d \geq 3$. $\square$

We next isolate the reconstruction step that occurs after the two inductive halves have become the same Dyck-prefix partition.

For a family $\mathcal{C} = \{C_i\}$ of nonempty subsets of $\{0, 1\}^{r+1}$, write

$$C_i^a = \{z \in \{0, 1\}^r : za \in C_i\}, \quad a \in \{0, 1\},$$

let $I(\mathcal{C})$ be the indices for which both C_i^0 and C_i^1 are nonempty, and define

$$\mathcal{G}_0(\mathcal{C}) = \{C_i^0 : C_i^0 \neq \emptyset\} \cup \{C_i^1 : i \in I(\mathcal{C})\},$$

$$\mathcal{G}_1(\mathcal{C}) = \{C_i^1 : C_i^1 \neq \emptyset\} \cup \{C_i^0 : i \in I(\mathcal{C})\}.$$

These are the two split families used in Shankar's induction.

Lemma 3.3 (reconstruction). *Let $r \geq 0$. Suppose a family $\mathcal{C}$ of nonempty subsets of $\{0, 1\}^{r+1}$ satisfies the three conditions of Theorem 3.1, and suppose*

$$\mathcal{G}_0(\mathcal{C}) = \mathcal{G}_1(\mathcal{C}) = \{\{z\} : z \in \{0, 1\}^r\}.$$

Then exactly one of the following holds:

$$(a) \ \mathcal{C} = \{\{z0\}, \{z1\} : z \in \{0, 1\}^r\};$$

$$(b) \ r \geq 1 \text{ and}$$

$$\mathcal{C} = \{\{t10, t01\} : t \in \{0, 1\}^{r-1}\}.$$

Proof. Every nonempty half C_i^a is a singleton, since it occurs in one of the two displayed singleton partitions. There is also no overlap between opposite halves when one of the corresponding blocks is split. Indeed, suppose $i \in I(\mathcal{C})$, $j \neq i$, and

$$z \in C_j^0 \cap C_i^1.$$

Choose $x \in C_i^0$. Then $z1, x0 \in C_i$, so condition (ii) gives $B_{r+1}(z1, x0) = r + 1 \pmod{2}$. Weight homogeneity gives $\text{wt}(x) = \text{wt}(z) + 1$, and (3) therefore yields

$$B_{r+1}(z0, x0) = B_{r+1}(z1, x0) = r + 1 \pmod{2}.$$

On the other hand, $B_{r+1}(z0, z1) = B_r(z, z) = r \pmod{2}$, contradicting constancy on $C_j \times C_i$. The case $C_j^1 \cap C_i^0 \neq \emptyset$ is symmetric.

It follows that each label $z \in \{0, 1\}^r$ either contributes the two singleton blocks $\{z0\}, \{z1\}$, or occurs in exactly one split block. Such a block has the form

$$\{u0, v1\}, \quad \text{wt}(u) = \text{wt}(v) + 1, \quad (17)$$

where the weight relation follows from condition (i). The labels used in distinct two-element blocks are disjoint.

Fix a block (17), and write $k = \text{wt}(v)$. If a label z is unpaired, cross-constancy with the two singleton blocks $\{z0\}$ and $\{z1\}$ gives

$$\Delta_{u,v}(z) = \mathbf{1}_{\{\text{wt}(z)=k+1\}} = \mathbf{1}_{\{\text{wt}(z)=k\}}. \quad (18)$$

If another two-element block is $\{a0, b1\}$, comparison of the two rows in its two columns gives

$$\Delta_{u,v}(a) = \mathbf{1}_{\{\text{wt}(a)=k+1\}}, \quad \Delta_{u,v}(b) = \mathbf{1}_{\{\text{wt}(b)=k\}}. \quad (19)$$

Equations (18)–(19) imply that $\Delta_{u,v}$ vanishes outside the two levels $k, k + 1$. Lemma 3.2 therefore says that every paired block is canonical,

$$\{t10, t01\},$$

except possibly for the unique alternating pair in odd dimension.

First exclude that exception. Suppose $r = 2s + 1 \geq 3$ and

$$u = 1(10)^s, \quad v = 0(01)^s.$$

By (12) and (18)–(19), every word of weight $s + 1$ must occur as a high endpoint and every word of weight s as a low endpoint. All other pairs are canonical. A canonical high endpoint ends in 1, while a canonical low endpoint ends in 0. If $s \geq 2$, there are $\binom{2s}{s-1} > 1$ weight- s words ending in 1, so one of them is different from v ; it can be neither the exceptional low endpoint nor a canonical low endpoint, a contradiction. If $s = 1$, the exceptional block is $\{1100, 0011\}$ and one forced canonical block is $\{0110, 0101\}$. Their B_4 -matrix is

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix},$$

contrary to cross-constancy. Thus no exceptional pair occurs.

It remains to show that the canonical pairs are either absent or all present. Suppose the pair indexed by $t \in \{0, 1\}^{r-1}$ is present, and put $h = \text{wt}(t)$. For $u = t1$ and $v = t0$, recurrence (3) gives

$$\Delta_{u,v}(z) = 1 \iff \text{wt}(z_1 \cdots z_{r-1}) = h. \quad (20)$$

No label of total weight h or $h + 1$ can be unpaired by (18). Combining (20) with (19) shows that the labels

$$s0 \text{ (wt}(s) = h), \quad s1 \text{ (wt}(s) = h - 1), \quad s1 \text{ (wt}(s) = h), \quad s0 \text{ (wt}(s) = h + 1)$$

must respectively be low, high, high, and low endpoints. Hence every canonical pair whose prefix weight lies in $\{h - 1, h, h + 1\} \cap \{0, \dots, r - 1\}$ is present. Iterating along the connected sequence of weight levels $0, 1, \dots, r - 1$ forces all canonical pairs. This proves the dichotomy. $\square$

Proof of Theorem 3.1. Proceed by induction on m , proving the inequality and its equality statement simultaneously. The case $m = 0$ is immediate. For every i , write

$$A_i^a = \{x \in \{0, 1\}^{m-1} : xa \in A_i\}, \quad a \in \{0, 1\},$$

and let I be the set of indices for which both A_i^0 and A_i^1 are nonempty. Consider the two collections of nonempty sets

$$\begin{aligned} \mathcal{G}_0 &= \{A_i^0 : A_i^0 \neq \emptyset\} \cup \{A_i^1 : i \in I\}, \\ \mathcal{G}_1 &= \{A_i^1 : A_i^1 \neq \emptyset\} \cup \{A_i^0 : i \in I\}. \end{aligned}$$

Each collection satisfies the three hypotheses in dimension $m - 1$. The weight condition is immediate, and (3) transfers internal and cross-constancy. It remains to verify disjointness. Suppose, for example, that

$$u \in A_j^0 \cap A_i^1$$

with $i \in I$. If $i = j$, then $u0, u1 \in A_i$ but $B_m(u0, u1) = m - 1$, contradicting (ii). Thus $i \neq j$. Choose $x \in A_i^0$. Then $u1, x0 \in A_i$, so $B_m(u1, x0) = m$. Since these two words have the same total weight, $\text{wt}(x) = \text{wt}(u) + 1$, and (3) gives

$$B_m(u0, x0) = B_m(u1, x0) = m.$$

On the other hand $B_m(u0, u1) = m - 1$. This contradicts constancy on $A_j \times A_i$. The other possible overlap is symmetric.

Put $a_i = |A_i^0|$ and $b_i = |A_i^1|$. By induction,

$$\sum_{i \notin I} a_i^2 + \sum_{i \in I} (a_i^2 + b_i^2) \leq 2^{m-1}, \quad (21)$$

$$\sum_{i \notin I} b_i^2 + \sum_{i \in I} (a_i^2 + b_i^2) \leq 2^{m-1}. \quad (22)$$

For $i \in I$,

$$(a_i + b_i)^2 \leq 2a_i^2 + 2b_i^2,$$

with equality exactly when $a_i = b_i$. Adding (21) and (22) proves (8).

Assume now that equality holds. Then both (21) and (22) are equalities and $a_i = b_i$ for every $i \in I$. By induction there are Dyck prefixes P, Q of length $m - 1$ such that $\mathcal{G}_0$ and $\mathcal{G}_1$ are their block partitions. Moreover B_{m-1} is constant on every block of $\mathcal{G}_0$ times every block of $\mathcal{G}_1$: if the two prefix blocks come from A_i^α and A_j^β , this follows from (3), since its correction term is constant on that product and B_m is constant on $A_i \times A_j$ (also when $i = j$). Proposition 2.4 yields $P = Q$.

Delete from every prefix word the coordinates belonging to $\mathcal{E}(P)$. Equivalently, apply Lemma 2.1 to the Dyck prefix PU , whose last position is unmatched; this preserves all B -conditions, including the final bit of the original words. Since every block of P collapses to one label in $\{0, 1\}^r$, the reduced reconstruction is exactly the situation of Lemma 3.3.

That lemma gives the new Dyck prefix PU in case (a) and PD in case (b), completing the induction and proving existence. Conversely, if P has k matched edges and r unmatched positions, Lemma 2.1 shows that its blocks satisfy (i)–(iii), and

$$\sum_{\varepsilon \in \{0,1\}^r} |\mathcal{A}_\varepsilon(P)|^2 = 2^r (2^k)^2 = 2^{2k+r} = 2^m.$$

Thus every Dyck-prefix partition is indeed an equality case.

For uniqueness, note that from the union $\Omega(P)$ one recovers the matching by

$$\{i, j\} \in \mathcal{E}(P) \iff x_i \neq x_j \text{ for every } x \in \Omega(P).$$

If i, j are not matched, their bits can be made equal by independently orienting their matching edges and choosing unmatched bits. Hence the equality partition determines P . $\square$

4 Rigidity on the Boolean cube

The final equality step produces a bijection of the unmatched labels. The next proposition shows that this bijection cannot be nontrivial.

Proposition 4.1 (cube rigidity). *Let $\phi : \{0, 1\}^r \rightarrow \{0, 1\}^r$ be a bijection satisfying*

$$\text{wt}(\phi(x)) = \text{wt}(x), \quad B_r(\phi(x), \phi(y)) = B_r(x, y) \quad (23)$$

for all $x, y \in \{0, 1\}^r$. Then ϕ is the identity.

Proof. For words of adjacent weights, define an intrinsic relation

$$u \sim v \iff \text{wt}(u) = \text{wt}(v) + 1 \text{ and } \Delta_{u,v}(z) = 0 \text{ outside the two levels } \text{wt}(v), \text{wt}(u).$$

Lemma 3.2 identifies all edges of this relation. They are the canonical edges

$$t1 \sim t0 \quad (t \in \{0, 1\}^{r-1}), \quad (24)$$

plus, when $r \geq 3$ is odd, the single exceptional edge

$$1(10)^{(r-1)/2} \sim 0(01)^{(r-1)/2}.$$

The relation is preserved by ϕ . Indeed,

$$\Delta_{\phi(u), \phi(v)}(\phi(z)) = \Delta_{u,v}(z)$$

for every z , and ϕ preserves Hamming-weight levels.

If r is even, (24) is therefore preserved as a set. If $r \geq 3$ is odd, the two endpoints of the exceptional edge are the only vertices of degree two in the relation graph; hence the exceptional edge is intrinsically distinguished, and after removing it the canonical matching is again preserved. The cases $r = 0, 1$ are immediate. Thus in all cases ϕ maps every canonical pair to a canonical pair. Weight preservation prevents the two endpoints from being exchanged, so there is a weight-preserving bijection $\psi : \{0, 1\}^{r-1} \rightarrow \{0, 1\}^{r-1}$ such that

$$\phi(t\alpha) = \psi(t)\alpha \quad (t \in \{0, 1\}^{r-1}, \alpha \in \{0, 1\}).$$

Since the last bits differ,

$$B_r(t0, s1) = B_{r-1}(t, s).$$

Using (23),

$$B_{r-1}(\psi(t), \psi(s)) = B_r(\phi(t0), \phi(s1)) = B_r(t0, s1) = B_{r-1}(t, s).$$

Hence ψ satisfies the same hypotheses in dimension $r - 1$. Induction gives $\psi = \text{id}$ and therefore $\phi = \text{id}$. $\square$

5 Extremal relations

Call a relation $R \subseteq \{0, 1\}^n \times \{0, 1\}^n$ *admissible* if

$$\text{wt}(x) = \text{wt}(y) \quad ((x, y) \in R) \quad (25)$$

and

$$B_n(x, x') = B_n(y, y') \quad \text{for all } (x, y), (x', y') \in R. \quad (26)$$

Shankar proved that every admissible relation has at most 2^n elements [1, Lemma 2.2]. Theorem 3.1 gives its complete equality case.

Proposition 5.1. *If $R \subseteq \{0, 1\}^n \times \{0, 1\}^n$ is admissible and $|R| = 2^n$, then there are Dyck prefixes P, Q of length n , with the same number r of unmatched positions, such that*

$$R = \bigcup_{\varepsilon \in \{0, 1\}^r} \mathcal{A}_\varepsilon(P) \times \mathcal{A}_\varepsilon(Q). \quad (27)$$

Proof. Form the bipartite graph whose edges are the pairs in R , and ignore isolated vertices. Let A_j and B_j be the left and right vertex sets of its connected components. If two left vertices x, x' have a common neighbor y , then (26) gives

$$B_n(x, x') = B_n(y, y) = n \pmod{2}.$$

An induction on graph distance along an alternating path shows the same for any two vertices in one A_j , and similarly in one B_j : if $x_0 - y_1 - x_1 - \dots - y_\ell - x_\ell$ is a shortest alternating path, then (26) gives $B_n(x_0, x_\ell) = B_n(y_1, y_\ell)$, and the latter pair has smaller distance. Also, (25) makes each A_j and B_j weight-homogeneous. Finally, if A_i and A_j are different components, then B_n is constant on $A_i \times A_j$. Indeed, if $x, x' \in A_i$ share a neighbor y and (a, b) is any edge of the other component, then

$$B_n(x, a) = B_n(y, b) = B_n(x', a).$$

Connectivity lets one move either endpoint, so the value is constant. The same holds on the right. Thus both collections satisfy Theorem 3.1.

The standard component estimate is

$$|R| \leq \sum_j |A_j| |B_j| \leq \left(\sum_j |A_j|^2 \right)^{1/2} \left(\sum_j |B_j|^2 \right)^{1/2} \leq 2^n. \quad (28)$$

Since $|R| = 2^n$, equality holds throughout. In particular every component is complete bipartite, and both quadratic sums equal 2^n . Theorem 3.1 gives Dyck prefixes P, Q whose block partitions are $\{A_j\}$ and $\{B_j\}$. If their numbers of unmatched positions are r_P, r_Q , then the common number of components is both 2^{r_P} and 2^{r_Q} , so $r_P = r_Q =: r$. Consequently P and Q also have the same number $k = (n - r)/2$ of matched edges.

Index the blocks of P by $\varepsilon \in \{0, 1\}^r$. The components give a bijection $\phi : \{0, 1\}^r \rightarrow \{0, 1\}^r$ such that

$$R = \bigcup_{\varepsilon \in \{0, 1\}^r} \mathcal{A}_\varepsilon(P) \times \mathcal{A}_{\phi(\varepsilon)}(Q).$$

A word in $\mathcal{A}_\varepsilon(P)$ has weight $k + \text{wt}(\varepsilon)$, and a word in $\mathcal{A}_{\phi(\varepsilon)}(Q)$ has weight $k + \text{wt}(\phi(\varepsilon))$. Completeness of the component and (25) therefore give

$$\text{wt}(\phi(\varepsilon)) = \text{wt}(\varepsilon).$$

For labels ε, η , choose arbitrary words in the two corresponding complete components. Equation (26), followed by Lemma 2.1, gives

$$B_r(\phi(\varepsilon), \phi(\eta)) = B_r(\varepsilon, \eta).$$

Proposition 4.1 gives $\phi = \text{id}$, proving (27). $\square$

6 Proof of the Catalan classification

Let $F = f_1 \cdots f_{2n}$ be a path from $(0, 0)$ to (n, n) , viewed as a balanced binary word, and define

$$u(F) = (f_1, \dots, f_n), \quad v(F) = (1 - f_{2n}, \dots, 1 - f_{n+1}). \quad (29)$$

Then $\text{wt}(u(F)) = \text{wt}(v(F))$. Conversely every pair (u, v) of equal weight reconstructs a unique balanced word by taking first half u and second half the reversed complement of v . Thus (29) is a bijection between paths from $(0, 0)$ to (n, n) and equal-weight pairs in $\{0, 1\}^n \times \{0, 1\}^n$.

Reversing and complementing the second half is a reflection of that half-path and preserves common edges. Therefore, for any two balanced paths F, G ,

$$B_{2n}(F, G) = B_n(u(F), u(G)) + B_n(v(F), v(G)) \quad \text{in } \mathbb{F}_2, \quad (30)$$

which is Observation 2.1 of [1].

Proof of Theorem 1.1. Let $\mathcal{F}$ be extremal and let

$$R = \{(u(F), v(F)) : F \in \mathcal{F}\}.$$

By (30), the even-intersection hypothesis says exactly that R is admissible for distinct pairs; for a pair with itself, both sides of (26) equal $n \pmod{2}$, so admissibility holds in full. Since $|R| = 2^n$, Proposition 5.1 gives Dyck prefixes P, Q of length n with r unmatched positions and

$$R = \bigcup_{\varepsilon \in \{0,1\}^r} \mathcal{A}_\varepsilon(P) \times \mathcal{A}_\varepsilon(Q).$$

Write the unmatched positions as

$$a_1 < \cdots < a_r \quad \text{in } P, \quad b_1 < \cdots < b_r \quad \text{in } Q.$$

Construct a matching M on $[2n]$ as follows:

- include every edge of $\mathcal{E}(P)$ in the left half;
- reflect every edge $\{i, j\} \in \mathcal{E}(Q)$ to $\{2n + 1 - j, 2n + 1 - i\}$ in the right half;
- for $1 \leq i \leq r$, add the cross-edge

$$\{a_i, 2n + 1 - b_i\}. \quad (31)$$

This is a perfect matching. It is noncrossing: internal edges are noncrossing; the right endpoints in (31) decrease as the left endpoints increase, so the cross-edges are nested; and an unmatched position of a Dyck prefix cannot lie strictly inside one of its matched edges, since it would still be on top of the stack when that edge closed.

For an internal edge of P , membership in a block $\mathcal{A}_\varepsilon(P)$ means that the corresponding two first-half bits are opposite. The same is true for the reflected edges of Q , because complementing both endpoint bits preserves opposition. Finally, the common label ε gives

$$f_{a_i} = v(F)_{b_i} = 1 - f_{2n+1-b_i},$$

so the two bits on each cross-edge (31) are opposite. Hence every $F \in \mathcal{F}$ belongs to $\mathcal{F}_M$. Both families have size 2^n , so

$$\mathcal{F} = \mathcal{F}_M.$$

The matching is unique. Indeed, for every noncrossing perfect matching M ,

$$\{i, j\} \in M \iff w_i \neq w_j \text{ for every } w \in \mathcal{F}_M. \quad (32)$$

The forward implication is the definition. If i and j lie on different matching edges, orient those two edges independently so that $w_i = w_j$. Thus (32) recovers M from the family. Since the noncrossing perfect matchings of $[2n]$ are counted by C_n , the number of extremal families is exactly C_n . $\square$

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